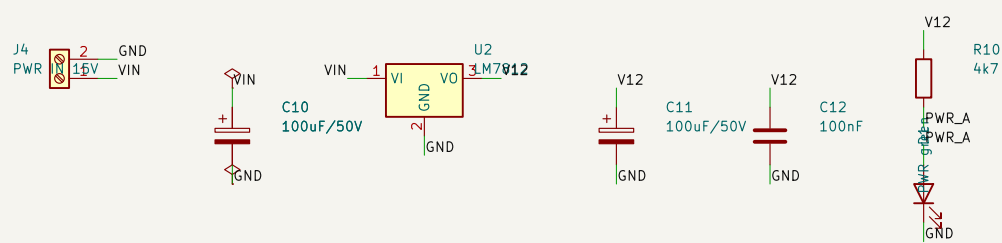


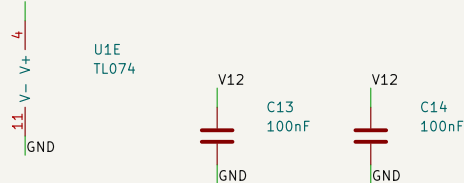
Bose Companion 5 sub crossover -- mono-summing 2nd-order Sallen-Key low-pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid-rail divider).

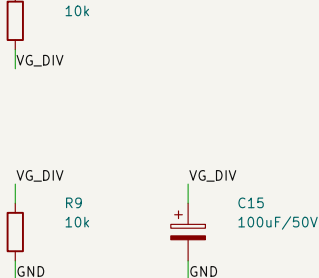
A POWER IN $\rightarrow +12\text{ V}$



B U1 SUPPLY

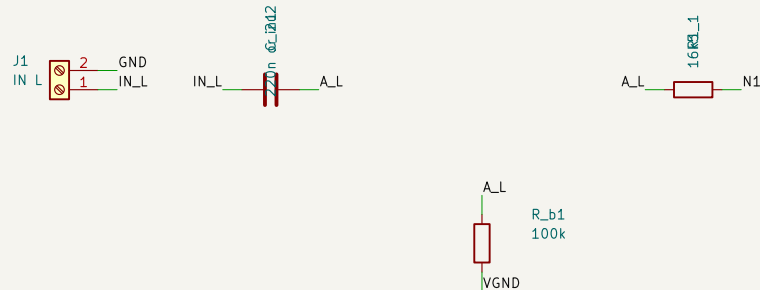


C VIRTUAL GROUND 6 V



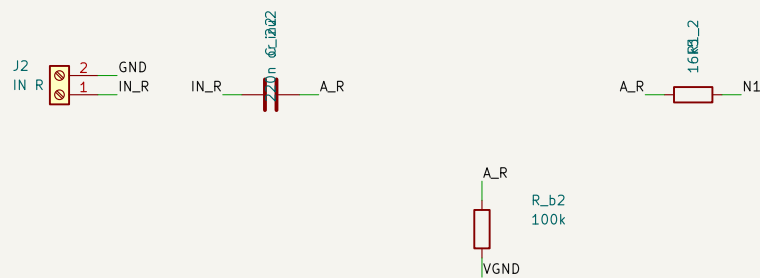
U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

D INPUT L

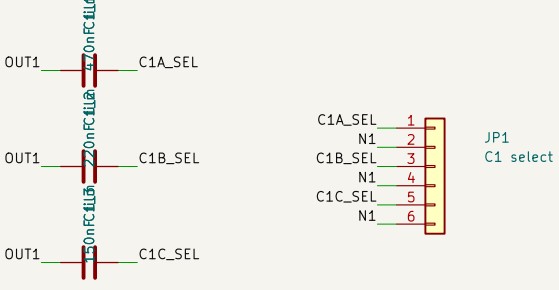


R_b1/R_b2 are the only DC path to A1's + input (C_in blocks DC, C2 is a capacitor). Without them A1 drifts to a rail. They sit after the coupling caps -- 100k at N1 would shift every corner by 8%.

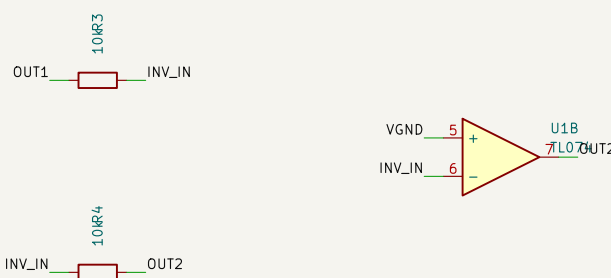
E INPUT R



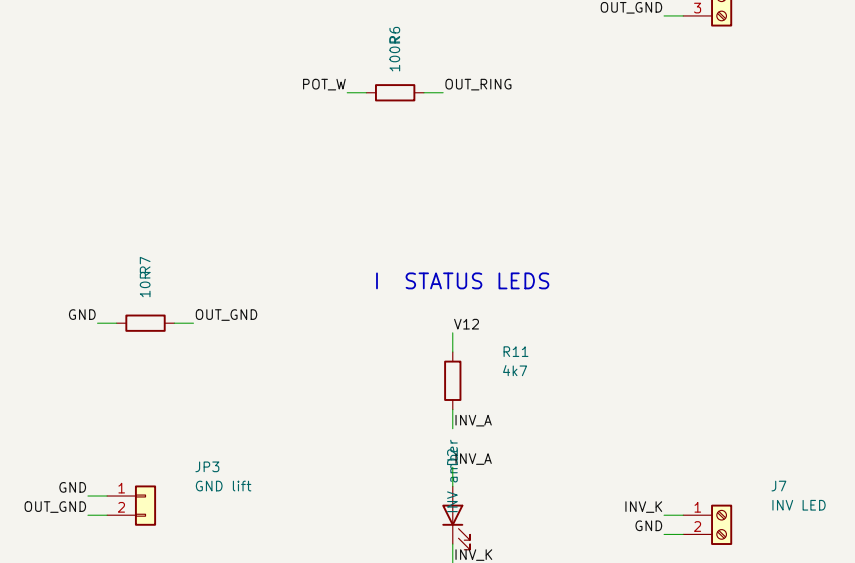
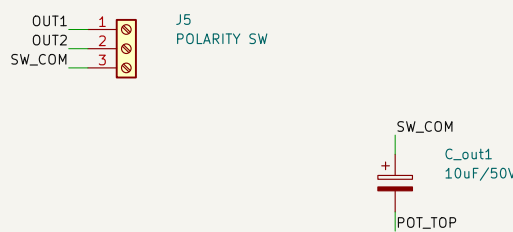
F Sallen-Key Low-Pass



G POLARITY INVERTER



H OUTPUT STAGE



C_{out} + faces SW_{COM}: that node sits at 6 V DC, the pot side at 0 V.
 J6 pin 3 = GND, not VGND — C_{out} has already removed the 6 V bias;
 referencing the pot to VGND would thump the driver at power-on.
 JP3 shorts R7 (ground lift). Normally fitted.

D1 = power lamp on V12 (proves the LM7812, not just VIN).
D2 = INVERTED lamp, fed from SW2's unused second pole via J7
Both ~2 mA. No LED touches a signal node.

C_{in1}/C_{in2} use a dual-pitch footprint: 5.00 mm for the 220n as built (or a 2u2 non-polarised electrolytic), 15.00 mm for a real 2u2 film part. Fit ONE part, 220n puts -8.8k of reactance in series with R1 at 82 Hz, so it is part of the filter, not just an input high-pass -- it costs 1.3-3.1 dB of passband level and damps the peaking. Whether that is a defect or a feature is a listening question; Gate 11 has not answered it, so both pitches are on the board and the envelope reserves room for the larger part either way.

Off-board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
 Process: single-sided PCB, bottom copper only, board outline 100 x 100 mm.
 Copper is isolation-milled on the CNC with an 0.8 mm flat end mill; the fiber
 laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
 The 0.8 mm tool width is a hard floor on every copper-to-copper gap. Fixed-pitch
 pads sit below the 0.85 mm target and carry a .kicad_dru exception: DIP-14
 0.94 mm (passes), 2.54 mm headers and the T0-220 0.84 mm (do not).
 Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c -> C1_1/C1_2/C1_3, C2a/C2b/C2c -> C2_1/C2_2/C2_3, R1a/R1b -> R1_1/R1_2, C_out -> C_out1.

Single-sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono-summing 2nd-order Sallen-Key low-pass, single +12 V rail

Sheet: /
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Title: Bose Companion 5 sub crossover

Size: A2	Date: 2026-08-16	Rev: B
KiCad: EDA 10.0.4		Id: 1/1